

HackerMate Developer Guide

Welcome to the team! This guide outlines the setup procedures, database integrations, and collaboration workflows for the HackerMate repository.

Step 1: Install Prerequisites

Ensure the following software packages are installed on your local environment:

- **Git:** Distributed version control system (for cloning & pushing).
- **Node.js (v20 or newer):** Next.js runtime environment (includes npm).
- **VS Code:** Suggested editor with ESLint and TypeScript tools configured.

```
# Verify your local environments
git --version
node -v
npm -v
```

Step 2: Clone the Repository

Clone the project repository to your workspace and navigate to the project directory:

```
git clone <GITHUB_REPO_URL>
cd HackerMate_Backup
```

Step 3: Install Dependencies

Restore the package node modules specified in the `package.json` manifest:

```
npm install
```

Step 4: Create Environment Variables

Create a local environment file named `.env.local` in the root directory. Paste the configuration secrets provided by the team administrator:

```
# Public Supabase Access
NEXT_PUBLIC_SUPABASE_URL=https://your-project.supabase.co
NEXT_PUBLIC_SUPABASE_ANON_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...

# Server-Side Configuration (Never expose to client)
SUPABASE_SERVICE_ROLE_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
CRON_SECRET=your_secure_cron_secret
RESEND_API_KEY=your_resend_api_key
```

Step 5: Apply Database Migrations

The messaging, matchmaking, and team collaboration features require custom Row-Level Security (RLS) policies, triggers, and RPC functions. These are maintained inside the `supabase/migrations/` folder.

Apply the migrations by linking the Supabase CLI directly to your project ref:

```
# 1. Link your local project directory to your Supabase Project
supabase link --project-ref <project-ref>

# 2. Push database schema, RPCs, and RLS policies
supabase db push
```

💡 **Database Backup Alert:** Always back up your active staging database before executing a database push. Migrations overwrite active security policies.

Step 6: Enable Google & GitHub Auth

HackerMate uses Google and GitHub OAuth providers for secure developer registration:

1. Open your **Supabase Dashboard** and navigate to **Authentication > Providers**.
2. Enable **Google** and **GitHub**.
3. Enter the Client IDs and Secrets generated from Google Developer Console and GitHub Settings.
4. Ensure the Redirect URL matches your local environment: `http://localhost:3000`.

Step 7: Seed Hackathon Data

To test hackathon search and teammate matching features locally, you need to populate the database tables with crawled hackathon information:

```
# Run the local SQL seed using Supabase Dashboard SQL Editor or terminal
psql -d "postgresql://postgres:[password]@db.[project-ref].supabase.co:5432/postgres" -f seed_unsto
```

Step 8: Spin Up the Server

Start the local Next.js development server to verify setup:

```
npm run dev
```

Open `http://localhost:3000` in your browser. Complete the user onboarding flow (skills, GitHub verification, and college selection) to confirm successful connection

Daily Development Workflow

Maintain repository consistency by executing these Git steps during daily development sessions:

```
# 1. Pull changes before beginning a session
git pull origin main

# 2. Compile and test changes locally
npm run lint
npm run build

# 3. Commit and push contributions
git add .
git commit -m "feat: descriptive message of your modifications"
git push origin main
```

Troubleshooting Guide

If `npm install` Fails

Clear your local lockfiles and node module caching, then retry installation:

```
rm -rf node_modules package-lock.json
npm install
```

If Server Fails to Launch

Double-check environment variables in `.env.local`. If Database errors occur, verify migrations are up-to-date.

OAuth Sign-in Redirect Failures

Ensure the callback URL configuration matches in the provider console (e.g. Google Cloud or GitHub OAuth developer app redirects).

Cron Reminders Blocked

Verify the `CRON_SECRET` header matches the token passed by the server request. Ensure `RESEND_API_KEY` is correctly loaded.

Core Development Rules

- 📄 • **Lint First:** Never push code that breaks `npm run lint`.
- **Coordinate Database Changes:** Coordinate with the backend team before pushing modifications in `supabase/migrations/`.
- **Mock Notifications:** In sandbox developer environments, email notifications redirect automatically to your sandbox recipient to prevent production spam.